Progressive Parsing and Commonality Distillation for Few-Shot Remote Sensing Segmentation

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***Abstract*— In recent years, few-shot segmentation (FSS) has received widespread attention from scholars by virtue of its superiority in low-data regimes. Most existing research focuses on natural image processing, and very few studies are dedicated to the practical but challenging topic of remote sensing image understanding. Related experimental results show that directly transferring the previously proposed framework to the current domain is prone to produce unsatisfactory results with *incomplete objects* and *irrelevant distractors*. Such phenomena can be attributed to the lack of modules specifically designed for the complex characteristics of remote sensing images, e.g., great intra-class diversity and low target-background contrast. In this article, we propose a conceptually simple and easy-to-implement framework to tackle the aforementioned problems. Specifically, our innovative design embodies two main aspects: 1) the support mask is progressively parsed into multiple valuable subregions that can be further exploited to compute local descriptors with segmentation cues about intractable parts; and 2) the base-class memories stored in the meta-training phase are replayed and leveraged for the distillation of novel-class prototypes, where the commonalities between classes are adequately explored, more in line with the concept of *learning to learn*. These two components, i.e., the progressive parsing module and commonality distillation module, contribute to each other and together constitute the proposed PCNet. We conduct extensive experiments on the standard benchmark to evaluate segmentation performance in few-shot settings. Quantitative and qualitative results illustrate that our PCNet distinctly outperforms previous FSS approaches and sets a new state-of-the-art.**

***Index Terms*— Few-shot learning, few-shot segmentation (FSS), image processing, remote sensing, semantic segmentation.**

I. Introduction

THE analysis and understanding of remote sensing images can be conducted from multiple levels and perspectives [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11].

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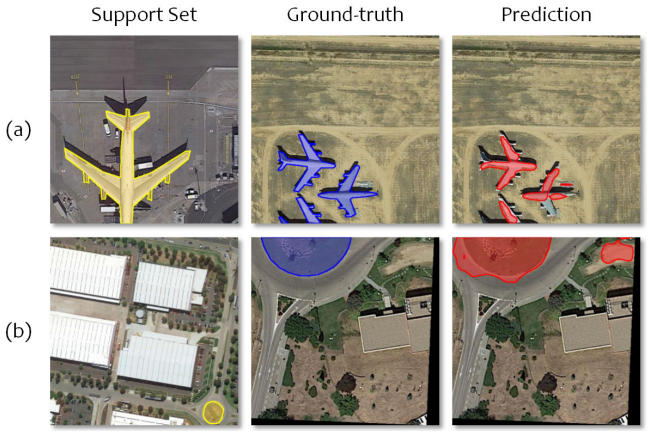


Fig. 1. **Segmentation results of advanced FSS approach on remote sensing images.** (Left to Right) Support set, ground-truth, and prediction. It can be noticed that previous frameworks fail to tackle the segmentation task of remote sensing images under the low-data regime, and tend to produce predictions with (a) *incomplete objects* and (b) *irrelevant distractors*.

One of the critical high-level processing techniques is semantic segmentation, which aims to discriminate the geospatial categories of all locations within the given image [12], [13], [14]. With the advancement of sensor devices and deep learning, remarkable technological breakthroughs have been achieved, driving applications in military defense, precision agriculture, environmental protection, and other fields [15], [16], [17]. However, there are two sides to almost everything, and even such promising technologies are no exception. Training satisfactory neural networks results in a serious waste of human and material resources, and the generalization of the derived model is highly fragile [18]. Specifically, 1) data annotation at this stage inevitably requires expert intervention [19], and tasks like segmentation that easily consume tens of thousands of densely annotated training samples are notoriously labor-intensive; and 2) the trained model produces output only about known classes and cannot be easily extended to prediction tasks for unseen classes. All these unfavorable factors hinder the further development of semantic segmentation.

In such a context, few-shot segmentation (FSS) has been proposed by scholars to handle dense prediction tasks in low-data regimes [20]. Shaban et al. [21], as the pioneers of FSS, first established the task setting and evaluation protocols. Meanwhile, they designed a deep network with a dual-branch structure, where the conditioning branch generates a set of classifier parameters embedded with category knowledge for

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the segmentation branch. Wang et al. [22] borrowed the idea of metric learning common in few-shot classification tasks to devise the method structure and introduced the prototype alignment regularization to enhance the discriminative nature of feature learning on this basis. Zhang et al. [23] developed a masked average pooling technique for the efficient utilization of the support mask, avoiding the excessive complexity of two separate backbone networks used in the traditional architecture. Given that implementing the fine-tuning strategy with very little data available is prone to overfitting, Zhang et al. [24] froze the backbone network during both training and inference stages to improve the model's ability to generalize to unseen categories. On the whole, impressive research progress has been made in the field of FSS over the years. Still, the vast majority of existing work only concerns the analysis and processing of natural images without further investigation into other more meaningful domains. For the remote sensing image segmentation targeted in this article, we have conducted certain exploratory experiments. Fig. 1 depicts some qualitative results of the advanced FSS method, and here [25] is selected for its superiority and representativeness. It can be observed that the previously well-performing framework does not give satisfactory results in the current setting; moreover, incomplete objects [e.g., plane in Fig. 1(a)] and irrelevant distractors [e.g., roundabout in Fig. 1(b)] often appear in the segmentation masks. We argue that this experimental phenomenon is closely related to the lack of modules specifically developed for the complex characteristics of remote sensing images, such as great intra-class diversity and low target-background contrast.

To alleviate the problems mentioned above and compensate for the research deficiency, we propose the progressive parsing module and commonality distillation module for the few-shot framework of remote sensing images, referred to as PCNet. The *former* module progressively parses the support mask into multiple valuable subregions, and a set of local descriptors is then computed to provide segmentation cues about intractable parts of geospatial objects; the *latter* module replays the memory units stored during the meta-training phase and distills abundant prototype representations by virtue of the commonality between the base class and the novel class. Compared with previously proposed methods, our PCNet enjoys a superior ability to discriminate error-prone foreground regions and derive holistic prototypes with rich category information. All these properties are highly beneficial for improving the recall of segmentation objects and suppressing irrelevant distractors. Our main contributions are as follows.

1) To tackle the problem of remote sensing scene segmentation in low-data regimes, we design a conceptually simple and easy-to-implement framework, known as PCNet.
2) The two proposed critical components, i.e., the progressive parsing module (PPM) and commonality distillation module (CDM), complement each other to remedy the frequent occurrence of incomplete objects and irrelevant distractors.
3) We perform extensive experiments on the standard dataset to verify the proposed framework, which achieves state-of-the-art performance in multiple settings.

II. Related Work

A. Generic Semantic Segmentation

The purpose of semantic segmentation is to identify the semantic class of each pixel within the given image. The fully convolutional neural networks (FCNs) proposed by Long et al. [26] for semantic segmentation can be regarded as a seminal work in the era of deep learning, with a profound impact on subsequent research. On the basis of FCNs, the studies related to semantic segmentation can be further divided into multiple subdirections. One line of research with a wide audience concentrates on the aggregation of multiscale contextual information. Chen et al. [27] devised the atrous spatial pyramid pooling module, which utilizes atrous convolution with different dilated rates to collaboratively capture and integrate multiscale features. Zhao et al. [28] proposed the pyramid pooling module that concatenates the feature representations obtained through pooling operations with different sizes of kernels and the original ones, allowing information interaction between different scales and different subregions. Another representative line of research is the acquisition of task-relevant representations through attention mechanisms [29]. In addition, some researchers are devoted to the semantic segmentation of remote sensing images [30], [31], [32]. Lv et al. [33] proposed a simple and practical segmentation network with spatial–spectral attention mechanism, and its effectiveness was demonstrated via ten quantitative evaluation metrics.

B. Few-Shot Learning

Few-shot learning is an important research direction that explores *how to cognize unseen concepts using a handful of annotated samples* [34], [35], [36], [37], [38]. Part of the research work, inspired by the idea of metric learning, aims to learn a discriminative high-dimensional space in which the features of the same category are close together, while the features belonging to different categories are far apart. Representative approaches include the matching network [39] that proposes an episodic training paradigm, the prototypical network [40] that utilizes prototypes to characterize categories, and the relation network [41] that performs classification via learnable metrics. Another part of the work starts from the perspective of model optimization and strives to design more effective fine-tuning strategies under few-shot settings. Finn et al. [42] obtained a set of initialization parameters that can be quickly adapted in a meta-learning manner, laying a good foundation for fine-tuning the few-shot model on unseen categories. Rusu et al. [43] proposed to learn the generative representation of model parameters in a low-dimensional latent space, which facilitates capturing uncertainty in the data and adapting more efficiently with sparse samples. There are also some related studies in the field of remote sensing image processing; however, most of them focus on scene classification tasks [44], [45].

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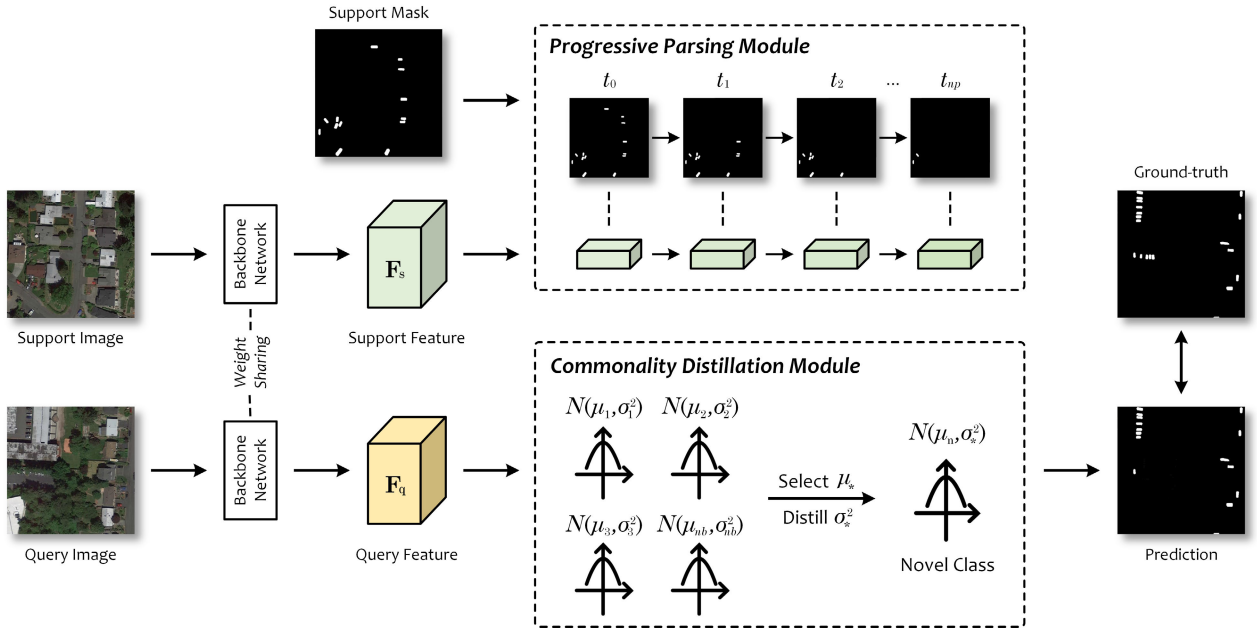


Fig. 2. **Architecture illustration of our PCNet**, including backbone network, PPM, and CDM. Given the support and query images, we first extract the intermediate features using a weight-sharing backbone network. Then, the support mask is progressively parsed based on the support features to obtain multiple valuable subregions that can be leveraged to compute local descriptors. Eventually, the base-class memory stored during the meta-training phase distills abundant novel-class prototypes for segmentation prediction according to interclass affinities.

C. Few-Shot Segmentation

FSS is an emerging research direction in computer vision, aiming at tackling dense prediction tasks with only scarce annotated data [20], [46]. As pioneers in the FSS field, Shaban et al. [21] took the lead in establishing task settings, standard benchmarks, and evaluation criteria in 2017, which facilitated the subsequent series of research. In recent years, benefiting from the focus on *few-shot*, a number of novel approaches have been proposed and achieved impressive performance. For example, the prototype learning network proposed by Dong and Xing [47] drew on the ideas of prototype learning and metric learning, where the prototype fusion and probability fusion modules make multiclass prediction possible. The prototype alignment network (PANet) proposed by Wang et al. [22] also tackled the FSS problem from a metric learning perspective, in particular by introducing prototype alignment regularization to facilitate knowledge mining from the support data. To obtain more robust support representations, Zhang et al. [23] proposed a masked averaged pooling strategy that includes only regions belonging to the support category for computation.

Nevertheless, most existing FSS frameworks are limited to natural image processing and do not explore their applications in other fields, such as remote sensing image interpretation. Yao et al. [48] leveraged multiple prototypes to perform detailed matching, exhibiting remarkable segmentation capability in aerial images. The deep metametric learning network (DMML-Net) proposed by Wang et al. [49] was optimized in a metametric training paradigm, which enjoys flexible scalability and can be rapidly generalized to novel episodes with a few gradient steps. In view of the lack of research in the current field, we specifically design two effective modules, i.e., the progressive parsing module and commonality distillation module, to boost the robustness of the previous approaches for handling remote sensing images, and accordingly integrate them into a unified few-shot framework termed PCNet.

III. Methodology

A. Task Description

FSS, as an extension of few-shot learning techniques in the field of image segmentation, seeks to determine the corresponding regions of the same category in the query image based on a small number of annotated images. Following the standard FSS setup [21], the original dataset is divided into two parts, the training set $\mathcal{D}_{\text{train}}$ and the test set $\mathcal{D}_{\text{test}}$, based on semantic categories $\mathcal{C}_{\text{train}}$ and $\mathcal{C}_{\text{test}}$, where $\mathcal{C}_{\text{train}} \cap \mathcal{C}_{\text{test}} = \varnothing$. Both $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{test}}$ are composed of a large number of *episodes* [39], each of which includes a support set $\mathcal{S} = \{(\mathbf{I}_s^{(k)}, \mathbf{Y}_s^{(k)})\}_{k=1}^{K}$ and a query set $\mathcal{Q} = \{(\mathbf{I}_q, \mathbf{Y}_q)\}$, totaling $K+1$ image-label pairs. During the meta-training phase, we randomly sample a series of episodes from $\mathcal{D}_{\text{train}}$ to train the model to learn the mapping $f{:}\{(\mathcal{S}, \mathbf{I}_q)\} \rightarrow \{\mathbf{Y}_q\}$; while during the meta-testing phase, the learned mapping f is employed to retrieve the corresponding regions in the query image $\mathbf{I}_q$ sampled from $\mathcal{D}_{\text{test}}$ with respect to $\mathcal{S}$. The ground-truth (query) mask $\mathbf{Y}_q$ is provided during training and not available during testing.

B. Method Overview

The overall architecture of our proposed PCNet is presented in Fig. 2, where the PPM and CDM are the innovations of this work. Given a query image $\mathbf{I}_q$ and a support image $\mathbf{I}_s$, two backbone networks with shared weights are used to extract their intermediate feature representations $\mathbf{F_q}$ and

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$\mathbf{F_s} \in \mathbb{R}^{C\times H\times W}$, respectively. The PPM utilizes the support mask $\mathbf{Y}_s$ and support features $\mathbf{F_s}$ to perform self-inference, i.e., to predict the ground-truth of the support image. The intractable regions within the mask are mined and resolved in an iterative manner, yielding multiple local descriptors that contain diverse segmentation information. The CDM is responsible for storing representative base-class memories in the meta-training phase and performing memory updates according to certain guidelines. In the meta-inference phase, the constructed memory bank is replayed and leveraged for the distillation of novel-class prototypes, facilitating the model fitting of unseen concepts. The distilled prototypes are finally integrated and fed into a decoder network with multiscale feature aggregation to produce accurate segmentation masks.

C. Progressive Parsing Module

The PPM module utilizes the support mask and the support features extracted from the backbone network to parse the target region into multiple valuable subregions in a progressive manner, thus providing a more representative prototype vector.

For the convenience of understanding, we introduce the module pipeline in the order of mask parsing, i.e., from t_0 to t_{np}. Specifically, given the backbone features $\mathbf{F}_s$ and the original mask $\mathbf{Y}_s$ of the support branch, the masked global average pooling (GAP) operation is first performed to obtain holistic category representations $\mathbf{p}_0$, as shown in Fig. 3. Here we give a general equation for obtaining the prototype vectors $\mathbf{p}_p$, which is also applicable to the computation of the subsequent parsed local descriptors

$$\mathbf{p}_p = \frac{\sum_{x,y} \mathbf{F}_s^{(x,y)} \mathbb{1}\left[\mathbf{Y}_{s;p}^{(x,y)} = 1\right]}{\sum_{x,y} \mathbb{1}\left[\mathbf{Y}_{s;p}^{(x,y)} = 1\right]} \tag{1}$$

where $\mathbb{1}(\cdot)$ is an indicator function that outputs 1 if the condition is met or 0 otherwise. (x, y) indexes the spatial locations $H \times W$ of the feature or mask. $p \in \{0, \ldots, \text{np}\}$ is utilized to index all the prototypes involved in the PPM module. $\mathbf{Y}_{s;0} = \mathbf{Y}_s$ indicates the original support mask.

Afterward, the derived prototype performs pixel-level affinity evaluation with the support features, where the cosine metric is employed to produce a floating-point probability map $\mathbf{M}_s$ taking values between 0 and 1

$$\begin{aligned}\mathbf{M}_{s;p'+1}^{(x,y)} &= \text{Sim}\left(\mathbf{p}_{p'}, \mathbf{F}_s^{(x,y)}\right) \\ &= \frac{\mathbf{p}_{p'} \cdot \mathbf{F}_s^{(x,y)}}{\|\mathbf{p}_{p'}\| \cdot \left\|\mathbf{F}_s^{(x,y)}\right\|}\end{aligned} \tag{2}$$

where $\text{Sim}(\cdot, \cdot)$ represents the function that assesses the similarity of the two vectors, and $p' \in \{0, \ldots, \text{np}-1\}$ indexes the prototypes involved in the affinity calculation during the parsing process.

The prediction for the current parsing can thus be acquired based on the probability map $\mathbf{M}_s$ and the preset threshold τ

$$\hat{Y}_{s;p''}^{(x,y)} = \mathbb{1}\left[\mathbf{M}_{s;p''}^{(x,y)} \geq \tau\right] \tag{3}$$

where the threshold τ is a hyperparameter that determines the prediction region of the model in terms of the confidence score, and the impact of its value on segmentation performance is analyzed in Table VI. $p'' \in \{1, \ldots, \text{np}-1\}$ indexes all the predictions resulting from np parsings. To mine intractable parts in the support image, we compare the prediction mask $\hat{Y}_s$ with the ground-truth mask $\mathbf{Y}_s$ and identify those under-segmented foreground regions, which can be defined as

$$\mathbf{Y}_{s;p''+1}^{(x,y)} = \mathbb{1}\left[\left(\mathbf{Y}_{s;p''}^{(x,y)} - \hat{Y}_{s;p''}^{(x,y)}\right) = 1\right] \tag{4}$$

where $\mathbf{Y}_{s;p''+1}^{(x,y)}$ denotes the newly acquired support mask, which also serves as a candidate mask to generate the prototype for the next parsing.

After the full execution of the parsing procedure, all the parsed prototypes $\{\mathbf{p}_1, \ldots, \mathbf{p}_{\text{np}}\}$ and the holistic prototype $\mathbf{p}_0$ are integrated through a convolution operation to yield more discriminative class descriptors, which can be formulated as

$$\mathbf{p}_s = \text{Conv}\left(\mathbf{p}_0 \oplus \mathbf{p}_1 \cdots \oplus \mathbf{p}_{\text{np}}\right) \tag{5}$$

where $\text{Conv}(\cdot)$ is a 1×1 convolution operation with the number of input channels and output channels being $C \times (\text{np} + 1)$ and C, respectively. $\oplus$ represents the concatenation operation along the channel dimension. $\mathbf{p}_s$ is employed as the support prototype for feeding into the follow-up CDM module.

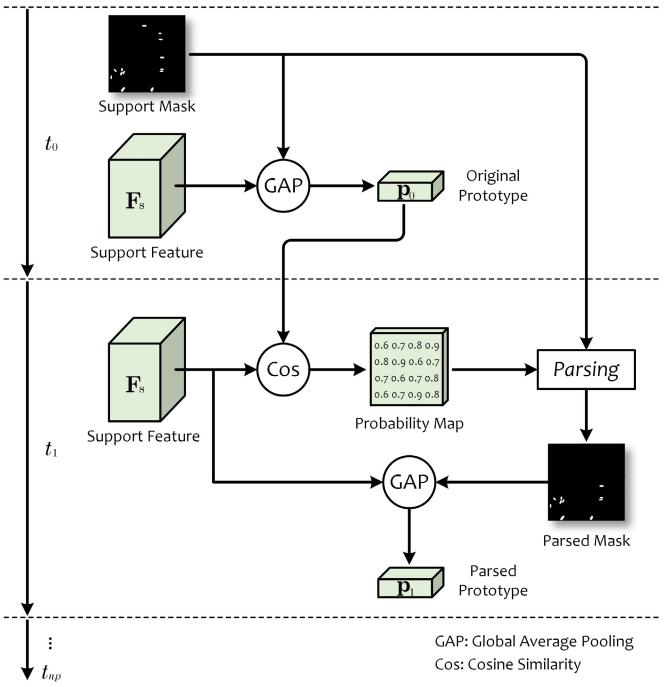


Fig. 3. **Illustration of the PPM.** Before the parsing starts (t_0), we obtain the original prototype of the foreground region utilizing the GAP technique. Then, the prototype is evaluated for cosine similarity with the support features during the parsing process. Based on the preset threshold τ and the previously parsed mask, the derived probability map is further mined for undersegmented (*false negatives*) essential regions. The newly acquired mask with the corresponding prototype is used for the next parsing until the maximum parsing number np is reached.

D. Commonality Distillation Module

The CDM module stores a particular scale of base-class memory $\mathbf{p}_b^{(i,j)}$ during the meta-training phase and

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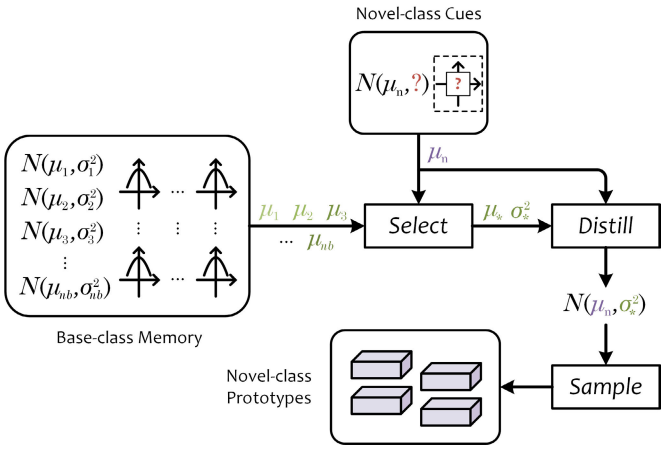


Fig. 4. **Illustration of the CDM.** During the *meta-training* phase, the base-class memory is stored and updated according to certain principles; during the *meta-testing* phase, the memory unit with the highest affinity score with the novel class is utilized for prototype distillation. In this way, the sparse novel-class segmentation cues are complemented to facilitate target identification under the few-shot paradigm.

replays it to distill more novel-class prototypes $\mathbf{p}_n^{(l)}$ during the meta-testing phase, as depicted in Fig. 4, where $i \in \{1, \ldots, \text{nb}\}$, $j \in \{1, \ldots, \text{nm}\}$, and $l \in \{1, \ldots, \text{ns}\}$ index the base class, the memory unit of each class, and the sampled novel prototype, respectively. nb, nm, and ns denote the number of base classes, the number of memories per class, and the number of sampled novel-class prototypes, respectively. Assuming that the memory of each base class obeys a normal distribution, the corresponding random variable can be mathematically represented as $Z_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$, where μ_i and σ_i represent the mean and standard deviation, respectively. Considering the *incompleteness* of novel-class representations under the few-shot paradigm and the intra-class *diversity* of remote sensing images, we perform prototype distillation using the most closely related base class, which can be formulated by

$$\star = \underset{i}{\text{Argmax}}(\text{Sim}(\mu_i, \mathbf{p}_s)) \tag{6}$$

where $\star \in \{1, \ldots, \text{nb}\}$ indicates the selected index. The standard deviation $\sigma_\star$ corresponding to the selected base class is then leveraged to construct the normal distribution of the novel class, i.e., $Z_n \sim \mathcal{N}(\mu_n, \sigma_\star^2)$. In this way, the base-class knowledge gained over a substantial number of annotated samples is transferred to the novel class, providing multiple robust representations $\{\mathbf{p}_n^{(l)}\}_{l=1}^{\text{ns}}$ for the tricky remote sensing segmentation task in few-shot settings.

For the distilled novel-class prototypes $\mathbf{p}_n^{(l)}$ and the integrated support prototype $\mathbf{p}_s$ output from the PPM module, we perform a weighted fusion based on the following equation:

$$\mathbf{p}_s^{\text{ref}} = \beta \mathbf{p}_s + (1 - \beta) \sum_l \alpha^{(l)} \mathbf{p}_n^{(l)} \tag{7}$$

where $\mathbf{p}_s^{\text{ref}}$ denotes the refined support prototype which is finally fed into the decoder network to provide the essential segmentation cues. β is the fusion coefficient for the original and sampled prototypes which is empirically set to 0.7; while α is the fusion coefficient of the sampled prototypes which can be set as a constant, e.g., $1/\text{ns}$, or can be calculated in an *adaptive* manner as follows:

$$\alpha^{(l)} = \frac{w^{(l)}}{\sum_l w^{(l)}} \tag{8}$$

where the fusion weight $w^{(l)}$ of each sampled novel-class prototype $\mathbf{p}_n^{(l)}$ can be mathematically defined as

$$w^{(l)} = 1 \Big/ \left(1 + \lambda_0 e^{-\gamma_0 v_s^{(l)}}\right) \times 1 \Big/ \left(1 + \lambda_1 e^{-\gamma_1 v_q^{(l)}}\right) \tag{9}$$

where λ_0, λ_1, γ_0, and γ_1 are empirically set to 2, 1, 5, and 5, respectively, according to [50]. v_s and v_q evaluate the affinity of each sampled prototype $\mathbf{p}_n^{(l)}$ with the support prototype $\mathbf{p}_s$ and query features $\mathbf{F}_q$, respectively, and can be expressed as

$$v_s^{(l)} = \text{Sim}\left(\mathbf{p}_n^{(l)}, \mathbf{p}_s\right) \tag{10a}$$

$$v_q^{(l)} = \frac{1}{HW} \sum_{x,y} \text{Sim}\left(\mathbf{p}_n^{(l)}, \mathbf{F}_q^{(x,y)}\right). \tag{10b}$$

With such parameter settings, the fusion between sampled novel-class prototypes $\{\mathbf{p}_n^{(l)}\}_{l=1}^{\text{ns}}$ comprehensively consider the support and query features and the affinity score with query features contributes more to the fusion weights $w^{(l)}$.

E. K-Shot Extension

The procedure of the method described above is mainly for one-shot tasks, and here we present the extension of our PCNet on K-shot segmentation tasks. Considering that there may be more than one base class suitable for prototype distillation of the novel class (e.g., ship versus small vehicle and large vehicle), the sampling of prototypes is performed independently when multiple support images are available. As a result, the number of prototypes being sampled also changes from ns to $K \times$ ns

$$\mathbf{p}_n^{(k,l)} = \varpi\left(\mu_n, \sigma_\star^{(k)}\right) \tag{11}$$

where $\varpi(\cdot, \cdot)$ indicates the sampling operation given the mean and standard deviation of the normal distribution. μ_n represents the mean of prototype representations $\{\mathbf{p}_s^{(k)}\}_{k=1}^{K}$ from multiple support samples.

F. Network Training

The proposed model is optimized in a mini-batch fashion, where each batch denotes a specific episode. To improve the effectiveness of the mask parsing process and facilitate the generation of more representative local descriptors, we impose an additional segmentation constraint on the PPM module, i.e., the progressive parsing regularization (PPR). Specifically, the integrated prototype $\mathbf{p}_s$ generated by (5) is evaluated for affinity with the support features $\mathbf{F}_s$ to obtain the probability map $\mathbf{M}_s$. We leverage this foreground-related map to construct a two-channel segmentation prediction and compare it with the ground-truth mask $\mathbf{Y}_s$ for loss evaluation

$$\mathcal{L}_{\text{PPR}} = \text{BCE}((1 - \mathbf{M}_s) \oplus \mathbf{M}_s, \mathbf{Y}_s) \tag{12}$$

where $\text{BCE}(\cdot, \cdot)$ stands for the binary cross entropy loss. $\oplus$ is the concatenation operation along channel dimensions.

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The overall loss function for model optimization can be mathematically expressed as follows:

$$\mathcal{L} = \mathcal{L}_{\text{seg}} + \eta \mathcal{L}_{\text{PPR}} \tag{13}$$

where $\mathcal{L}_{\text{seg}}$ is the primary segmentation loss regarding the query image. η is the balancing coefficient set to 0.5 since different values bring slight performance improvement.

IV. Experiments

A. Experimental Setup

1) Datasets: Currently, there are various kinds of research on natural image FSS; however, their extension to remote sensing image segmentation is still minimal. For a fair comparison, we follow the previous work [48] to construct a few-shot benchmark based on the iSAID dataset [51] that includes 6 55 451 finely annotated object instances across 15 geospatial categories in 2806 high-resolution images. In this work, these semantic categories are equally divided into three splits to serve as unseen categories under the few-shot paradigm respectively, as presented in Table I. The segmentation model is meta-trained on two splits and meta-tested on the remaining split in a cross-validation fashion.

2) Evaluation Metrics: Following previous work in the field [52], we adopt the mean intersection over union (mIoU) as our primary evaluation metric and the foreground-background IoU (FB-IoU) as a secondary metric. Specifically, the equations are as follows: $\text{mIoU} = (1/\text{nn})\sum_{c=1}^{\text{nn}} \text{IoU}_c$, where nn denotes the number of unseen classes in each target fold, and IoU_c is the IoU of class c; $\text{FB-IoU} = (1/2)(\text{IoU}_F + \text{IoU}_B)$, where the two terms on the right side of the equation represent the foreground and background IoU.

3) Implementation Details: Referring to the mainstream FSS models [24], [25], [53], we use pre-trained backbone networks and freeze their parameters to boost the generalization ability. VGG-16 [54] and ResNet-50 [55] serve as backbones for a fair comparison with other methods. These backbones are initialized using weights pre-trained by the base-learner [53], while other layers are initialized by the PyTorch framework's default settings. Our proposed few-shot model is trained on the base dataset under the meta-learning paradigm [39], where the batch size and training epoch are set to 8 and 12, respectively. The stochastic gradient descent (SGD) optimizer is employed in our experiments, where the momentum and weight decay are set to 0.9 and 0.0001, respectively. The raw images from the iSAID dataset [51] are cropped to 512 × 512 and then used for model training. During the meta-testing phase, we randomly select 1000 episodes for performance evaluation, and the results presented are the average of five independent runs with different seeds. Our network is built under the PyTorch framework, and all experiments are run on NVIDIA GeForce RTX 3090 GPUs.

B. Comparison With State-of-the-Arts

We compare and analyze our model with state-of-the-art FSS approaches in this section. The superiority of the model is verified by in-depth evaluation from both quantitative and qualitative aspects. In Table II, the mIoU results for different methods under one-shot and five-shot settings are presented.

TABLE I

Unseen Categories for Each Split of iSAID-5^i Dataset

Split ID	Unseen categories
0	Ship; Storage tank; Baseball diamond; Tennis court; Basketball court
1	Ground track field; Bridge; Large vehicle; Small vehicle; Helicopter
2	Swimming pool; Roundabout; Soccer ball field; Plane; Harbor

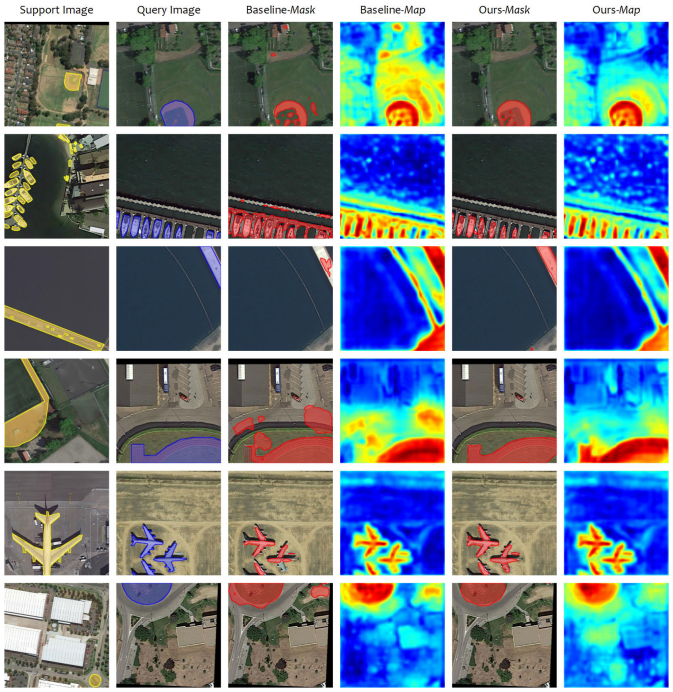


Fig. 5. **Qualitative results** of the proposed PCNet and baseline approach (from left to right: support image, query image, segmentation mask for baseline, activation map for baseline, our segmentation mask, and our activation map). Here the activation map is derived from the two-channel tensor of the network output. It can be observed that our PCNet significantly improves the *recall* of target segmentation (e.g., first, third, and fifth rows) and also performs better in terms of distractor suppression (e.g., second, fourth, and sixth rows) compared to the baseline framework.

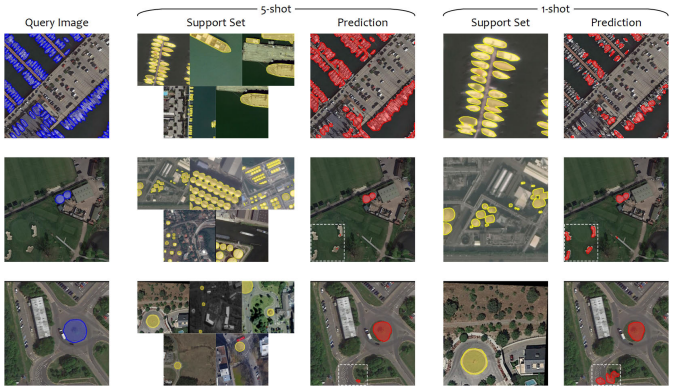


Fig. 6. **Qualitative results** of the proposed PCNet under one-shot and five-shot settings. We select one of the five-shot support images as the one-shot support image for comparison. The white dotted rectangles indicate the regions with significant differences in segmentation masks. Thanks to the richness of support information, the model discriminates targets more accurately and suppresses interferers more effectively.

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TABLE II

QUANTITATIVE RESULTS OF ONE-SHOT AND FIVE-SHOT SEGMENTATION ON iSAID-5^i USING MIOU METRIC. "#PARAMS." INDICATES THE NUMBER OF LEARNABLE PARAMETERS OF THE NETWORK. THE BEST PERFORMANCE IS SHOWN IN **BOLD**

Method	1-shot				5-shot				#Params.
	Split-0	Split-1	Split-2	Mean	Split-0	Split-1	Split-2	Mean	
VGG-16									
PANet [ICCV19] [22]	26.86	14.56	20.69	20.70	30.89	16.63	24.05	23.86	14.7M
CANet [CVPR19] [24]	13.91	12.94	13.67	13.51	17.32	15.07	18.23	16.87	14.2M
SCL [CVPR21] [56]	25.75	18.57	22.24	22.19	35.77	**24.92**	32.70	31.13	11.5M
NERTNet [CVPR22] [57]	25.78	**20.01**	19.88	21.89	38.43	24.21	28.99	30.54	20.5M
DCP [IJCAI22] [58]	28.17	16.52	22.49	22.39	39.65	22.68	29.93	30.75	10.6M
PFENet [TPAMI22] [25]	28.52	17.05	18.94	21.50	37.59	23.22	30.45	30.42	**10.4M**
DMML [TGRS22] [49]	24.41	18.58	19.46	20.82	28.97	21.02	22.78	24.26	14.7M
SDM [TGRS22] [48]	24.52	16.31	21.01	20.61	26.73	19.97	26.10	24.27	18.6M
*Ours-**PCNet***	**32.48**	19.88	**24.56**	**25.64**	**41.09**	21.98	**34.14**	**32.40**	10.9M
ResNet-50									
PANet[ICCV19] [22]	27.56	17.23	24.60	23.13	36.54	16.05	26.22	26.27	23.6M
CANet [CVPR19] [24]	25.51	13.50	24.45	21.15	29.32	21.85	26.91	26.03	22.3M
SCL[CVPR21] [56]	34.78	22.77	31.20	29.58	41.29	25.73	**37.70**	34.91	11.9M
NERTNet[CVPR22] [57]	34.93	23.95	28.56	29.15	44.83	26.73	37.19	36.25	20.8M
DCP[IJCAI22] [58]	37.83	22.86	28.92	29.87	41.52	28.18	33.43	34.38	11.3M
PFENet[TPAMI22] [25]	35.84	23.35	27.20	28.80	42.42	25.34	33.00	33.59	**10.8M**
DMML[TGRS22] [49]	28.45	21.02	23.46	24.31	30.61	23.85	24.08	26.18	23.6M
SDM[TGRS22] [48]	27.96	21.99	27.82	25.92	28.50	25.23	31.07	28.27	29.3M
*Ours-**PCNet***	**40.24**	**24.64**	**31.31**	**32.06**	**45.31**	**28.19**	37.36	**36.95**	11.3M

TABLE III

QUANTITATIVE RESULTS OF ONE-SHOT AND FIVE-SHOT SEGMENTATION ON iSAID-5^i USING FB-IOU METRIC. THE BEST PERFORMANCE IS **BOLDED**

Method	VGG-16		ResNet-50	
	1-shot	5-shot	1-shot	5-shot
PANet [ICCV19] [22]	55.65	54.51	56.56	57.37
CANet [CVPR19] [24]	53.15	54.82	56.64	59.46
SCL [CVPR21] [56]	59.10	62.81	61.30	64.13
NERTNet [CVPR22] [57]	59.58	63.25	59.97	64.45
DCP [IJCAI22] [58]	58.68	62.70	62.36	63.77
PFENet [TPAMI22] [25]	57.89	62.73	60.09	63.25
DMML [TGRS22] [49]	54.37	56.99	57.78	58.26
SDM [TGRS22] [48]	57.01	59.88	59.58	59.90
*Ours-**PCNet***	**59.89**	**63.48**	**62.64**	**65.12**

TABLE IV

ABLATION STUDIES OF THE PROPOSED COMPONENTS. "PPM" AND "CDM" DENOTE THE PROGRESSIVE PARSING MODULE AND COMMONALITY DISTILLATION MODULE, RESPECTIVELY

PPM	CDM	Split-0	Split-1	Split-2	Mean
✗	✗	28.52	17.05	18.94	21.50
✓	✗	31.89	19.22	23.72	24.94
✗	✓	30.59	18.90	23.62	24.37
✓	✓	**32.48**	**19.88**	**24.56**	**25.64**

TABLE V

ABLATION STUDIES OF THE MAXIMUM PARSING NUMBER np. "FPS" DENOTES THE AVERAGED FRAME-PER-SECOND

np	Split-0	Split-1	Split-2	Mean	FPS
0	30.59	18.90	23.62	24.37	**23.92**
1	31.69	18.94	**24.77**	25.13	23.38
3	32.02	19.32	22.54	24.63	22.47
5	**32.48**	19.88	24.56	**25.64**	21.10
7	31.37	**20.21**	**24.77**	25.45	20.45
10	31.24	19.11	24.09	24.81	18.83

TABLE VI

ABLATION STUDIES OF THE THRESHOLD τ FOR DETERMINING THE PARSED SEGMENTATION MASK

τ	Split-0	Split-1	Split-2	Mean
0.3	31.34	19.69	24.15	25.06
0.5	31.57	19.36	24.31	25.08
0.7	**32.48**	**19.88**	**24.56**	**25.64**
0.9	30.55	19.63	24.18	24.79

TABLE VII

ABLATION STUDIES OF THE NUMBER OF MEMORY (CAPACITY) PER CLASS IN TOTAL BASE-CLASS MEMORY. "#PARAMS." DENOTES THE NUMBER OF ADDITIONAL PARAMETERS INTRODUCED

nm	#Params.	mIoU	FB-IoU	FPS
1	**2,560**	24.28	58.82	**22.12**
3	7,680	24.53	58.75	21.65
5	12,800	**25.64**	**59.89**	21.10
10	25,600	25.06	59.07	20.58

It can be noticed that the proposed PCNet yields the best overall mIoU scores in all cases and also performs at a high level in each split. Particularly, our approach achieves performance gains of 3.25% and 2.19% over state-of-the-art FSS models under the one-shot setting with VGG-16 and ResNet-50 backbone networks, respectively, implying its unique advantages on the current task. Furthermore, we observe that the improved strategies do not introduce too many learnable parameters and remain comparable with advanced methods, e.g., 10.4 versus 10.9 M. The FB-IoU results shown in Table III also illustrate the effectiveness of the proposed framework.

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TABLE VIII

ABLATION STUDIES OF THE UPDATING STRATEGIES FOR THE BASE-CLASS MEMORY. "QUEUE" INDICATES THAT THE CLASS DESCRIPTOR OF THE CURRENT EPISODE REPLACES THE ONE THAT ENTERED THE MEMORY BANK FIRST. "ADAPTIVE" MEANS THAT UPDATES TO THE MEMORY BANK ARE PERFORMED BASED ON AFFINITY SCORES

Method	1-shot		5-shot	
	mIoU	FB-IoU	mIoU	FB-IoU
Queue	24.91	59.12	30.88	62.47
Adaptive	25.64	59.89	32.40	63.48

TABLE IX

ABLATION STUDIES OF THE NUMBER OF NOVEL-CLASS PROTOTYPES SAMPLED IN THE CDM

ns	Split-0	Split-1	Split-2	Mean	FPS
1	32.04	19.07	24.19	25.10	**21.59**
3	**32.48**	**19.88**	24.56	**25.64**	21.10
5	31.84	19.25	**24.64**	25.24	20.58
10	32.42	19.05	25.35	25.61	19.21

TABLE X

ABLATION STUDIES OF THE FUSION APPROACH OF THE ORIGINAL PROTOTYPE AND SAMPLED PROTOTYPES

Method	Split-0	Split-1	Split-2	Mean
Fixed-coefficient	32.25	18.23	24.04	24.84
Affinity-guided	32.48	19.88	24.56	25.64

Fig. 5 presents the segmentation results of the proposed PCNet with the baseline approach under the one-shot setting. It can be found that incomplete objects and irrelevant distractors often appear in the segmentation mask of the baseline approach, while our model exhibits superior target recall and interference suppression capabilities. Fig. 6 compares the segmentation results of our method in the one-shot and five-shot settings. Benefiting from the continuous mining of the support images and the effective utilization of the base-class knowledge, the performance of the proposed model is significantly improved with more samples available, as indicated by the white dotted rectangles in the figure.

C. Ablation Studies

Generally speaking, we adopt the VGG-16 backbone network for ablation studies, and the results presented are mainly the averaged mIoU metrics over three splits in one-shot and five-shot settings, unless otherwise specified.

1) Effects of the Proposed Modules: We first perform a module-level ablation study, which is crucial for analyzing the effectiveness of the improved scheme, as summarized in Table IV. The ultimate goal of either the PPM or CDM module is to provide a more representative prototype representation for the decoder network. However, the problems they tackle are different, with the former primarily focusing on those intractable object parts while the latter mainly concerning intra-class diversity. From the quantitative results, we observe that both modules consistently boost the accuracy of the few-shot model in all cases, and the variant with the PPM module performs slightly better, e.g., 24.94% versus 24.37% mIoU score.

2) Effects of the Maximum Parsing Number np: The maximum parsing number np determines the complexity of the PPM module. The larger the np, the more prototypes are parsed, and the more complex the module is. We take the values of np from {0, 1, 3, 5, 7, 10} to explore its effect on our proposed PCNet. It can be seen from Table V that np $= 5$ and np $= 7$ achieve the overall best and second best segmentation accuracy, while the variant with np $= 0$ naturally obtains the lowest complexity, i.e., 21.59 FPS. We apply np $= 5$ to the final scheme by a trade-off between precision and efficiency.

3) Effects of the Parsing Threshold τ: The parsing threshold τ in (3) can be viewed as a hyperparameter of the classifier, controlling the regions classified as foreground or background. In Table VI, it can be noticed that $\tau = 0.7$ exhibits the best performance, indicating that appropriately increasing the threshold for screening positive predictions is conducive to improving the quality of parsed regions.

4) Number of Memory per Class nm: The base-class memory stored by the CDM module during the meta-training phase contains a substantial amount of semantic information, the capacity nm of which also has certain impacts on our model. Table VII summarizes the performance metrics of the proposed PCNet in various aspects for different values of nm. We note that nm $= 5$ yields the optimal results for the segmentation metrics, i.e., 25.64% for mIoU and 59.89% for FB-IoU, while the larger capacity does not lead to further gains. The possible reason is that redundancy and confusion are created within the base-class memory, which in turn has a negative impact on the prototype integration [see (7)].

5) Effects of Updating Strategy for Base-Class Memory: For the base-class memory with a given capacity of nb $\times$ nm, it is essential to design updating strategies to cope with the semantic representations generated by new episodes. The common strategy is to update the memory bank as a queue, i.e., first-in-first-out and last-in-last-out. Such a strategy is highly effective at the beginning of model optimization since the newly yielded prototypes tend to have stronger representational capabilities. Nevertheless, when the model is close to convergence, the update of the memory bank will become unreasonable, and valuable base-class features may be overlooked. To this end, we design an adaptive updating strategy where the memory with the lowest affinity to the whole will be removed in each iteration. In this way, the intra-class compactness of the memory bank can be improved even after the model converges. Refer to Table VIII to compare the segmentation performance of these two schemes.

6) Number of Sampled Novel-Class Prototypes ns: Sampling multiple novel-class prototypes using the distillation strategy [see (6)] can be approximated as increasing the number of support data (*shots*), but not absolutely the more the better. Blindly increasing the sampling size will introduce excessive computational complexity, as an adaptive fusion [see (8)] is required within the distilled prototypes. Meanwhile, the risk of confusion between classes is raised, especially with base classes that provide distillation information. As shown in Table IX, the variant with ns $= 3$ yields impressive

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segmentation performance in low-data regimes, achieving the best overall mIoU of 25.64% and the second-best FPS of 21.10.

7) Effects of the Prototype Fusion Approach: As discussed in Section III-D, the fusion between the support prototype and sampled prototypes can be performed by either fixed-coefficient or affinity-guided approaches. The latter comprehensively considers the correlation between the prototype, support features, and query features for weight assignment, which is more suitable for few-shot tasks with significant differences between support-query image pairs. The quantitative results in Table X also validate the above views.

V. Conclusion

The great intra-class diversity and low target-background contrast of remote sensing images create inconvenience for the research of segmentation tasks in low-date regimes. For this reason, we specially designed two modules, the progressive parsing module, and the commonality distillation module, and integrated them into a unified framework called PCNet. The former continuously mines the support masks to provide multiple local descriptors that contain essential segmentation cues about intractable parts, while the latter leverages the base-class memory stored during the meta-training phase to distill abundant novel-class prototypes for prediction bootstrapping. We performed numerous ablation studies on the standard benchmark iSAID-5^i to verify the effectiveness of our approach and established new state-of-the-arts in the vast majority of experimental settings.

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